Systems of Equations

Week 1 Slide Deck

Math AI Course

What Is a System of Equations?

A system of equations is a group of two or more equations that use the same variables.

The goal is to find values of the variables that make all equations true at the same time.

For example:

$$\begin{cases} x - y = 5 \\ 2x + y = 7 \end{cases}$$

Solution of a System

A solution to a system of equations is usually written as an ordered pair.

For two variables, the solution has the form:

$$(x, y)$$

For example, if (x=4) and (y=-1), then the solution is:

$$(4, -1)$$

This means both equations are true when (x=4) and (y=-1).

Substitution Method

The substitution method means solving one equation for one variable and substituting it into the other equation.

From:

$$x - y = 5$$

we can solve for (x):

$$x = y + 5$$

Then substitute (x=y+5) into the second equation.

Example Using Substitution

Given the system:

$$\begin{cases} x - y = 5 \\ 2x + y = 7 \end{cases}$$

From the first equation:

$$x = y + 5$$

Substitute into the second equation:

$$2(y + 5) + y = 7$$

Solving the Example

Simplify:

$$2y + 10 + y = 7$$
$$3y + 10 = 7$$
$$3y = -3$$
$$y = -1$$

Then:

$$x = y + 5 = -1 + 5 = 4$$

So the solution is:

$$\boxed{(4, -1)}$$

Elimination Method

The elimination method solves a system by adding or subtracting equations to remove one variable.

For example, if one equation has $(+y)$ and another has $(-y)$, adding the equations can eliminate (y).

This method is useful when the coefficients are already opposites or can be made opposites.

Types of Solutions

A system of linear equations can have:

- **One solution**: the lines intersect at one point
- **No solution**: the lines are parallel
- **Infinitely many solutions**: the equations represent the same line

These outcomes help describe the relationship between the equations.

Summary

Systems of equations help us solve multiple equations with the same variables at the same time.

Key ideas:

- A solution must satisfy all equations.
- The **substitution method** replaces one variable with an equivalent expression.
- The **elimination method** removes one variable by adding or subtracting equations.
- A system can have **one solution**, **no solution**, or **infinitely many solutions**.

Example solution:

$$\boxed{(x, y) = (4, -1)}$$